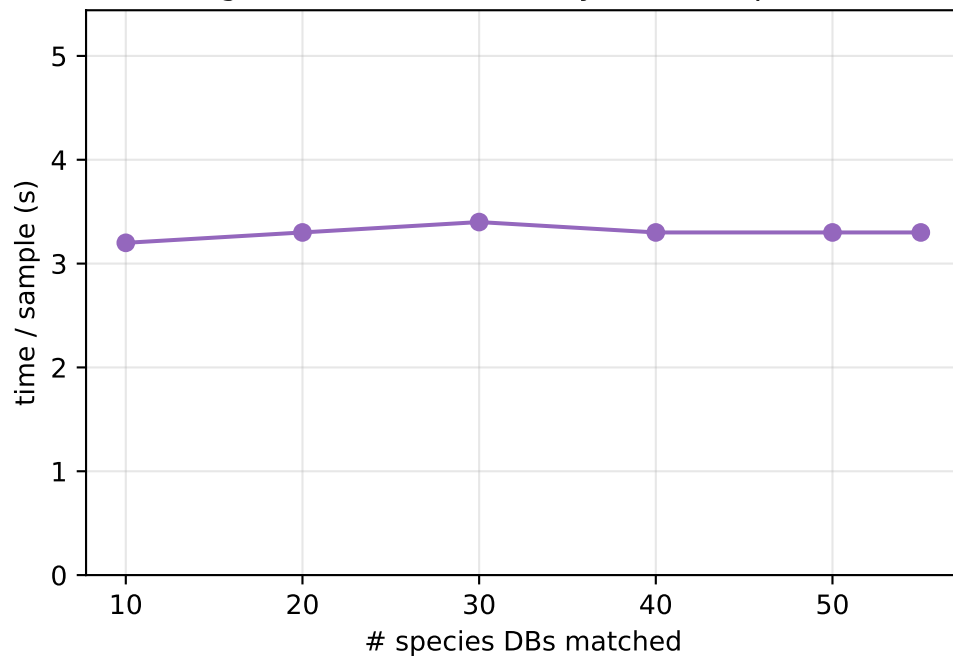
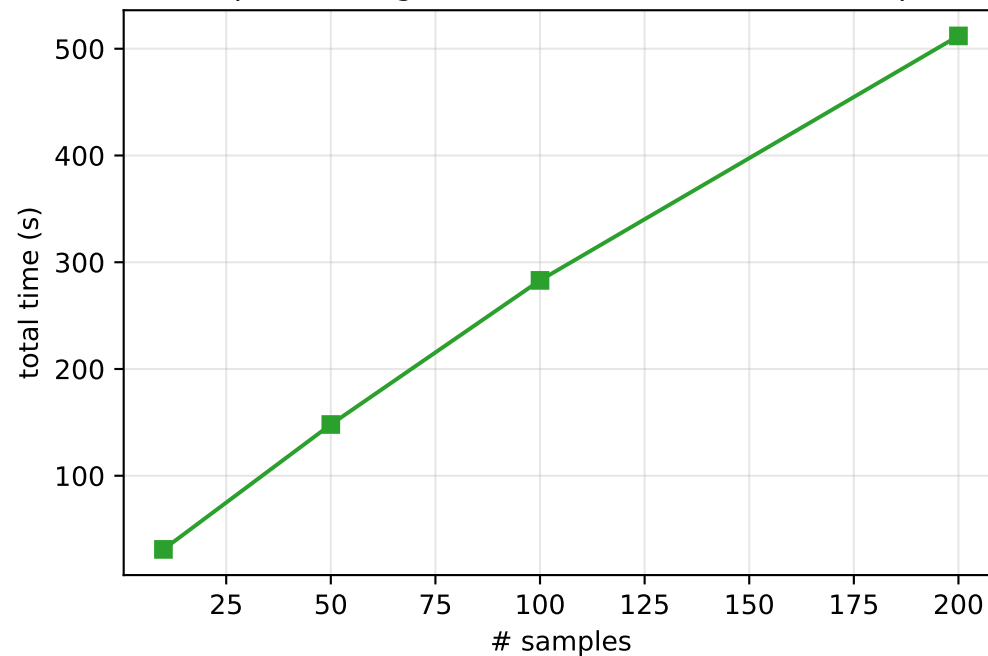


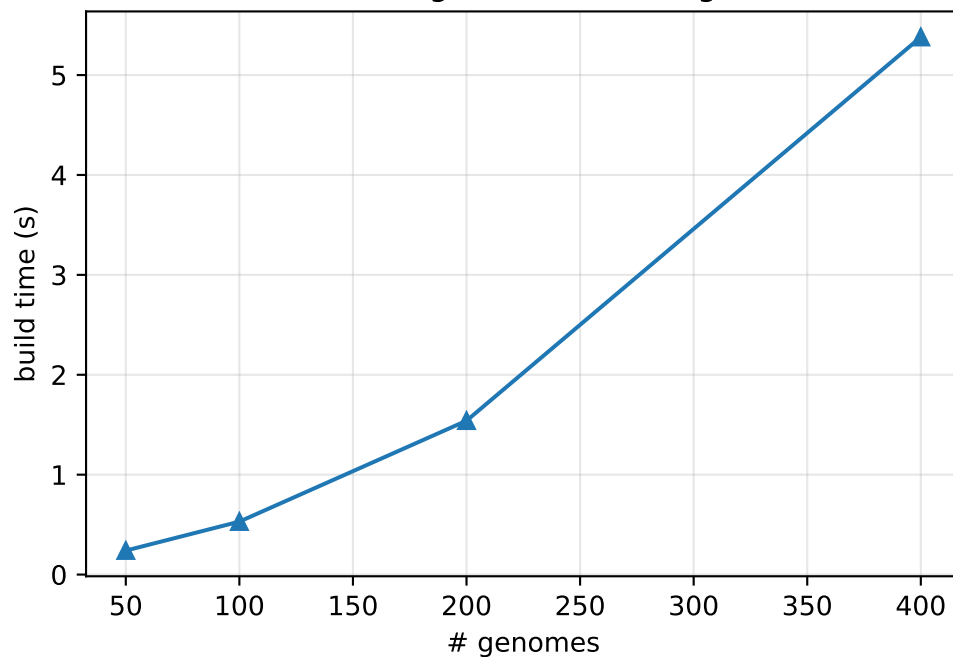
Species-count gradient — flat
(digest once, match many; 10→55 species)



Sample-count gradient — linear (~2.56 s/sample)



DB-build scaling (MinHash >96 genomes)



Multi-species accuracy vs species gate
(precision 0.96→1.0, recall 0.98)

